Unit-1

Lecture-1

Topic Name- Introduction on Data Structure

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Dtata Structure: Data structure means how to represent data in memory.

Data structure is defined as a logical and mathematical model for the organization of data.

Data structure tells about how to organize and stored data in memory, how to access data in memory i.e. access methods and what relationships data elements have among themselves.

Data structure is widely used in many aspects of computer science i.e. operating system, Graphics, Artificial intelligence, Algorithms etc.

Examples of Data Structure: Array, Stack, Queue, Linked –List, Tree, Graph.

Basic Terminology of Data Structure:

1. **Data:** Data is defined as elementary value or collection of values. Example: Employee's name, id are the data about the Employee.
2. **Data Item:** Data item refers to a single unit of value.
3. **Group Item:** Data item that are further divided into sub item is called as Group Item. For example: Student Name: First name+ Middle Name + Last Name.
4. **Record:** Collection of various data items is called Record. For example: student's name, address, id are grouped together to form record of student.
5. **File:** Collection of various records of one single type of entity is called Files.

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6. **Entity:** An entity is something that has some properties or attributes which may be assigned some numeric and non-numeric values. Example: Student, Employee.
7. **Attribute**: An attribute defines the property of Entity.
8. **Entity Set**: Entities having similar attributes form entity set.

Advantages of Data Structure:

1. **Abstraction:** Data Structure provides a level of abstraction. Data structures are also known as ADT (Abstract Data type).
2. **Efficiency:** Efficiency of Program depends upon the choice of right Data structure.
3. **Reusability:** Once Data structure is implemented it can be reused in other place.

Operations on Data Structure:

Operations on Data structure are as follows:

i. **Create**: Create operation is used to allocate memory for the data elements.
ii. **Insertion**: Insertion operation is used to add new data in a existing data structure.
iii. **Deletion**: Deletion operation is used to delete data from a data structure.
iv. **Searching**: Searching operation is used to search a specified data element in a data structure.
v. **Traversing**: Traversing operation is used to process all the data elements in a data structure.
vi. **Sorting**: Sorting means to arrange all the data elements in a data structure in some specified order.
vii. **Merging**: Merging means to combine elements of two similar data structure in order to form a new data structure of the same type.

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